HackAgon - Weekly Project Report

Team summary for each week, 06 Jul 2026 to 06 Oct 2026. Each week has the same fields as the Weekly Project Report Submission form: Week, GitHub / Report Link, Work Completed This Week, Problems Faced, Plan for Next Week.

Week 1 | 06 Jul - 10 Jul 2026 | Audit and decisions	
GitHub / Report Link	https://github.com/ASR-C/HACK-AGON
Work Completed This Week	**HITEN SHARMA:** Audited the accounts code (routes/auth.js, lib/auth.js): password hashing, JWT expiry, cookie flags. Drafted the room and match data model. **HARSH SHARMA:** Tested the judge (lib/judge.js) with Java, Python and C++ programs; confirmed time and memory limits and how each verdict maps. **HEMANTH SINGH TANWAR:** Chose the database host, reviewed schema.sql against the PRD, removed fake data from seed files, set up the Render staging service with a health check. **ARJUN SUMAN:** Tested tab-change, minimize, fullscreen-exit and camera capture in Chrome, Edge and Firefox. Wrote the proctor API contract. **AKSHAT RATHORE:** Collected reference screenshots, made Arena design previews, picked a direction with the team, wrote the style guide and first tokens.
Problems Faced	The code uses Node, Express and MySQL but the PRD says Java and JSP, so the documents had to be updated. MySQL versus Render's PostgreSQL needed a decision. Fake demo data was found in the seed files. Camera and fullscreen events behave differently across browsers.
Plan for Next Week	Finish OTP and profile; judge end to end; freeze schema V1 and deploy to staging; proctor endpoints; components and auth page styling.

Week 2 | 13 Jul - 17 Jul 2026 | Accounts, judge and base UI	
GitHub / Report Link	https://github.com/ASR-C/HACK-AGON
Work Completed This Week	**HITEN SHARMA:** Finished email OTP (5 minute expiry, 5 attempts, 60 second resend), added SMS OTP, forgot password, and profile edit that asks for a new OTP. **HARSH SHARMA:** Built code submit with language and question id, ran every test case, stored verdict, time and memory. Built question create, read and update API. **HEMANTH SINGH TANWAR:** Froze schema V1, finished the migration runner and repo layer, added smoke.js checks, set up automatic staging deploy from main. **ARJUN SUMAN:** Built snapshot upload, violation log and lock-state endpoints, chose snapshot storage, added duel and team tables with HEMANTH SINGH TANWAR. **AKSHAT RATHORE:** Chose fonts and tokens, built components.css (buttons, cards, chips, timer, form fields, empty state, toast), base layout and navigation, styled login, signup and verify-otp.
Problems Faced	No SMS (MSG91) client existed, so one had to be added. Mapping judge verdicts for runtime errors needed care. Schema changes needed coordination between members. Deciding where to store snapshots took discussion.
Plan for Next Week	Room creation API; admin screens and starter question bank; security middleware; duel and team APIs; create-room and Arena design.

Week 3 | 20 Jul - 24 Jul 2026 | Rooms and question bank	
GitHub / Report Link	https://github.com/ASR-C/HACK-AGON
Work Completed This Week	**HITEN SHARMA:** Built the room create API with validation and the PRD's editable defaults, wired create-room.html to it, and added server-run room status changes (open, live, finished). **HARSH SHARMA:** Built admin screens to add questions and test cases, loaded starter questions (arrays, strings, linked lists, stacks, queues, trees) and the filter API for the question side panel. **HEMANTH SINGH TANWAR:** Added rate limits on login and OTP, input validation, CORS rules, one error handler, request logs, and tests for auth routes. **ARJUN SUMAN:** Built duel invite and waiting-queue API, create and join team APIs for duo and squad, and team size rules. **AKSHAT RATHORE:** Designed create-room with the side question panel and filters, Arena room cards (not a plain grid), and empty states with personality.
Problems Faced	Writing test cases for every starter question took longer than planned. Room status had to be driven by a server timer rather than the browser. CORS rules blocked some requests while the frontend was being wired. Team size rules for duo and squad needed clarifying.
Plan for Next Week	Join and play flow; judge hardening; scoreboard queries and backups; duel and team screens; match page.

Week 4 | 27 Jul - 31 Jul 2026 | First playable match	
GitHub / Report Link	https://github.com/ASR-C/HACK-AGON
Work Completed This Week	**HITEN SHARMA:** Built join by Arena or invite code, start match, and the submission endpoint that calls the judge with a server-side on-time check. **HARSH SHARMA:** Limited concurrent judge runs, allowed one pending submission per player per question, tested known good and bad solutions per language, stored code for similarity checks. **HEMANTH SINGH TANWAR:** Added scoreboard queries and indexes, turned on backups, deployed Milestone 1 to staging and ran the test checklist. **ARJUN SUMAN:** Connected duel.html to the duel API; built team join and the teammate list inside a room. **AKSHAT RATHORE:** Built match.html: editor, question list, timer, polling scoreboard and the countdown animation.
Milestone	**Milestone 1 reached: two users play a full solo room on staging, and a new account shows zero everywhere.**
Problems Faced	The judge slowed down when many submissions arrived together. Late submissions near the end of the timer needed a server-side check. The staging checklist exposed a few small bugs that were fixed before the milestone.
Plan for Next Week	Results and profile counters; admin panel; money tables and ledger; duo, squad and duel play; My mistakes.

Week 5 | 03 Aug - 07 Aug 2026 | Results, teams and duels	
GitHub / Report Link	https://github.com/ASR-C/HACK-AGON
Work Completed This Week	**HITEN SHARMA:** Added final ranks with a tie-break rule, results page data, and profile counters calculated from real records. **HARSH SHARMA:** Added admin login protection, users list with block and unblock, question and book management; kept make-admin.js protected and removed fix-admin.js from deployment. **HEMANTH SINGH TANWAR:** Created tournaments, prize_splits, payments and an append-only wallet_ledger tables, opened the Razorpay test account, wrote the money API contract. **ARJUN SUMAN:** Made team score count each question once, automatic duel pairing, the platform daily room and always-open practice duels. **AKSHAT RATHORE:** Saved wrong submissions automatically, built the My mistakes list with last attempt, failing case and retry button, and its empty state.
Problems Faced	The tie-break rule for equal scores needed to be decided. An append-only ledger means corrections must be new entries. The play.js hook for saving wrong submissions had to be agreed between HITEN SHARMA and AKSHAT RATHORE. Fake counters had to be removed so everything starts at zero.
Plan for Next Week	Tournament mode; similarity checks; host funding and webhooks; webcam and tab lock; proctoring screens and book library.

Week 6 | 10 Aug - 14 Aug 2026 | Tournament mode and funding	
GitHub / Report Link	https://github.com/ASR-C/HACK-AGON
Work Completed This Week	**HITEN SHARMA:** Added the tournament mode option to room creation, the consent page before joining, and made the match page refuse locked players on every request. **HARSH SHARMA:** Compared submissions inside a room and produced a flag list for the host dashboard and admin. **HEMANTH SINGH TANWAR:** Built the Razorpay order for the prize pool and a webhook that verifies the signature and ignores repeated events; registration stays closed until the pool is funded. **ARJUN SUMAN:** Added camera permission and face check, snapshot every 20 to 30 seconds, tab, minimize and fullscreen events that set the lock on the server, 30 second network grace period and the locked screen. **AKSHAT RATHORE:** Designed the consent, camera check and lock screens; built the free book library page grouped by subject with licence shown.
Problems Faced	Webhooks can arrive twice, so repeats had to be ignored safely. The similarity threshold needed tuning so honest similar answers are not flagged. Short network drops caused false locks until the grace period was added.
Plan for Next Week	Elimination and age rule; disputes; entry fees and refunds; host dashboard; achievements.

Week 7 | 17 Aug - 21 Aug 2026 | Host controls, entry fees, achievements	
GitHub / Report Link	https://github.com/ASR-C/HACK-AGON
Work Completed This Week	**HITEN SHARMA:** Built host eliminate with a reason, blocked rejoin and removed eliminated players from ranking, and added the 18+ rule for paid tournaments. **HARSH SHARMA:** Built the player dispute form that blocks payout and the admin screen to review and decide. **HEMANTH SINGH TANWAR:** Built entry fee checkout and join after payment, refunds for cancelled tournaments and tournaments under the minimum players, and data for the rules page. **ARJUN SUMAN:** Built the host dashboard: live player list, camera status, violation counts, snapshot viewer, eliminate button, and the job that deletes snapshots after 30 days. **AKSHAT RATHORE:** Built the achievements table with editable thresholds, metrics from real counts, progress bars, unlock animation, and AI hints through routes/ai.js.
Problems Faced	Refund cases (cancelled versus under minimum players) needed care to avoid double refunds. The age check had to be enforced on the server, not only in the form. The host dashboard had to update live without heavy load.
Plan for Next Week	Result locking; payout admin tools; payout details and payouts; real proctoring trial; money screens.

Week 8 | 24 Aug - 28 Aug 2026 | Results lock and payouts	
GitHub / Report Link	https://github.com/ASR-C/HACK-AGON
Work Completed This Week	**HITEN SHARMA:** Built host confirm results and 24 hour auto-confirm with no dispute, and sent the final rank list to the money module. **HARSH SHARMA:** Built the failed payouts screen with retry, and review tools for flagged cheating and eliminated players. **HEMANTH SINGH TANWAR:** Built payout details with OTP and Razorpay account validation, and the payout job that deducts the 1% fee, splits by rank, handles leftover paise and checks the ledger balance. **ARJUN SUMAN:** Ran a 10-person test match, counted false locks, tuned the grace time and fixed problems on different browsers. **AKSHAT RATHORE:** Built the tournaments list, tournament-details rules page, checkout, payout status and winner screens.
Milestone	**Milestone 2 reached: one full paid tournament in test mode - host funds, players pay, cheaters locked, results confirmed, winners paid by the split, ledger balanced.**
Problems Faced	Splitting prizes into whole paise left remainders that needed a fixed rule. Payout failures needed retry handling. The 10-person trial showed a few false locks on certain browsers.
Plan for Next Week	Organization hosting; final admin dashboard; security review; duel polish; AI hints and design pass.

Week 9 | 31 Aug - 04 Sep 2026 | Organizations and security	
GitHub / Report Link	https://github.com/ASR-C/HACK-AGON
Work Completed This Week	**HITEN SHARMA:** Built the organization host application form, admin approval, higher limits for approved organizations, and finished the profile page. **HARSH SHARMA:** Built the final admin dashboard (disputes, flagged cheating, payout failures) and added medium and hard questions for every topic. **HEMANTH SINGH TANWAR:** Checked every route uses parameterized queries, tested injection, session fixation and token misuse, fixed findings, locked make-admin.js and fix-admin.js in production. **ARJUN SUMAN:** Built rematch and duel history, fed duel results to achievements and mistakes, and fixed problems from the proctoring trial. **AKSHAT RATHORE:** Built the AI hint after a failed attempt with daily limits per user, and ran a design consistency pass on the first half of the pages.
Problems Faced	The security tests found a few routes that needed stricter validation. Writing correct test cases for medium and hard questions is slow. Groq limits mean AI hints must be capped per user per day.
Plan for Next Week	End-to-end tests; judge load test; load test and email provider; proctoring edge cases; accessibility and copy.

Week 10 | 07 Sep - 11 Sep 2026 | Testing and load	
GitHub / Report Link	https://github.com/ASR-C/HACK-AGON
Work Completed This Week	**HITEN SHARMA:** Wrote a script that runs signup, room creation, join, solve and results; fixed what broke; did keyboard and screen-reader checks of auth and room pages. **HARSH SHARMA:** Ran a judge load test with 200 queued submissions, tuned concurrency, tested huge inputs and infinite loops, and confirmed limits are enforced. **HEMANTH SINGH TANWAR:** Simulated 200 to 500 players in one room, added indexes and a short scoreboard cache, and moved email from Gmail to a transactional provider. **ARJUN SUMAN:** Tested refresh, re-login, two tabs, unplugged camera and slow network; confirmed locks survive and honest players are not punished. **AKSHAT RATHORE:** Checked contrast (at least 4.5:1), added focus states, checked phone width on every main page, and rewrote leftover generic copy.
Problems Faced	Scoreboard polling was slow at high player counts until indexes and caching were added. Gmail limits made a transactional email provider necessary. The screen-reader check found a few missing labels.
Plan for Next Week	Legal pages; bug bash; production setup; mock matches; landing page and guides.

Week 11 | 14 Sep - 18 Sep 2026 | Legal pages, production and bug bash	
GitHub / Report Link	https://github.com/ASR-C/HACK-AGON
Work Completed This Week	**HITEN SHARMA:** Wrote terms of service, privacy policy, refund policy and webcam consent text, and added the acceptance checkbox at signup. **HARSH SHARMA:** Tested the app as a stranger would, logged and fixed judge and admin bugs, and confirmed the seed files contain only reference data. **HEMANTH SINGH TANWAR:** Set up a separate production service and database, wrote the Razorpay live-mode checklist, filed the written confirmation about paid contests, ran the backup restore test and set up error alerts. **ARJUN SUMAN:** Ran 3 mock matches with friends, fixed the bugs found, and wrote a short host guide for tournament mode. **AKSHAT RATHORE:** Built the final index page with real content, added short onboarding hints, took screenshots and wrote the demo script.
Problems Faced	Razorpay live mode needs the written confirmation about paid contests to be kept on file. Mock matches revealed a few small proctoring bugs. Legal text had to match what the app actually does.
Plan for Next Week	Code freeze; deploy to production; one small real tournament; live proctoring watch; demo preparation.

Week 12 | 21 Sep - 25 Sep 2026 | Launch	
GitHub / Report Link	https://github.com/ASR-C/HACK-AGON
Work Completed This Week	**HITEN SHARMA:** Code freeze, then watched signups, OTP delivery and login errors during the first real tournament. **HARSH SHARMA:** Handled user reports and reviewed flagged cheaters and disputes from the first event. **HEMANTH SINGH TANWAR:** Deployed to production, ran one small real tournament with friends, and checked every payout and the ledger afterwards. **ARJUN SUMAN:** Monitored locks and snapshots during the event and fixed urgent proctoring problems. **AKSHAT RATHORE:** Fixed the last visual issues, prepared the demo and presentation, and collected feedback from the first players.
Milestone	**Milestone 3 reached: production is live and one real tournament finished with correct payouts.**
Problems Faced	A few urgent issues came up during the live event and were fixed quickly. First-player feedback showed some small screens that were confusing.
Plan for Next Week	Fix issues from feedback; update guides and documents; final review and submission.

Week 13 | 28 Sep - 06 Oct 2026 | Wrap-up and final review	
GitHub / Report Link	https://github.com/ASR-C/HACK-AGON
Work Completed This Week	**HITEN SHARMA:** Reviewed signup and login errors from the first tournament, fixed small account and room bugs, re-ran the end-to-end test, updated api-contract.md. **HARSH SHARMA:** Reviewed remaining user reports and flagged cheating, closed open disputes, added more questions, re-ran judge checks for every language. **HEMANTH SINGH TANWAR:** Checked the ledger after the first tournament, reviewed Razorpay webhook logs, verified backups restore, updated deployment notes. **ARJUN SUMAN:** Reviewed proctoring logs, tuned lock rules to reduce false locks, re-tested duels and teams, updated the host guide. **AKSHAT RATHORE:** Fixed small UI issues from feedback, re-checked all pages on phone and desktop, updated the style guide, demo script and screenshots, did a final accessibility check.
Problems Faced	Only small issues remained, mostly from first-player feedback. Time was needed to finish documents and the final report.
Plan for Next Week	Submit the final report and project; use the retro list to plan the next features.